Learning a Smoother Quadratic Feedforward + PI Controller

An uncertainty-aware ringing penalty, its derivation, and nonlinear T-MATS validation

Computational study in the supplied T-MATS workspace

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Abstract

The preceding PILCO-style controller reduced speed-error RMSE but produced much larger fuel-command reversals. This study modifies policy learning to penalize expected excess total variation of fuel. A previous-command state and an event-start anchor retain the correlations needed to evaluate that penalty under Gaussian moment propagation. Seven coarse penalty weights and three intermediate refinements are optimized with the unchanged GP dynamics model and checked on the nonlinear gas-turbine plant. A declared joint tracking/ringing score selects a policy using only 10% load tests, after which a 5% test evaluates the frozen gains. The method remains an offline PILCO-style adaptation with numerical policy gradients and a frozen quadratic feedforward map. The report separates probabilistic predictions, plant measurements, and limitations.

1 Outcome and why the objective changed

The selected penalty is $\lambda = 0.2$, with $K_p = 0.02479762$ and $K_i = 0.05385920$. In the nonlinear 10% test, ringing falls by 85.9% relative to the previous PILCO policy, from 14.8252 to 2.0838 lbm/s. Speed RMSE changes from 1.1973 to 2.3877 rpm. The smoother command therefore has a measurable tracking cost.

The previous learned gains were $K_p = 0.06460568$, $K_i = 0.17102610$, $K_d = 0$. Their training objective rewarded small speed error and modest fuel magnitude, but neither term distinguished a smooth fuel trajectory from one repeatedly reversing around its equilibrium. A fast, aggressive PI policy could therefore score well despite oscillation. This was visible in the previous full nonlinear 10% test: RMSE was 1.19728 rpm and ringing was 14.82523 lbm/s, compared with 1.54614 rpm and 1.29904 lbm/s for retuned Poly2 with derivative feedback.

The present change acts in *policy learning*. No post-processing is applied to plant traces or video curves, and no extra low-pass filter is inserted into the fuel command. The deployed controller remains quadratic feedforward plus ordinary speed-error PI; changing its optimized gains changes the actual nonlinear plant response.

2 The experiment and what is held fixed

The reference is 9250 rpm at sea level, ISA zero and Mach zero. There are 60 s of preparation and 90 s of evaluation at $h = 0.015$ s. The three 10% shaft-load pulses occupy evaluation intervals $[10.005, 15)$, $[30, 40.005)$ and $[60, 75)$ s. Their magnitude is 1537.67919 hp, obtained from the same 15376.79195 hp trim as the preceding studies. Validation uses 768.83960 hp with identical timing. The numerical solver, speed sensor, fuel limits, feedforward coefficients and startup procedure are unchanged.

The dynamics GP is loaded from the previous study without refitting. Its training comprises 4242 transitions from the first two pulse windows of base PI, original Poly2 and retuned Poly2. A 40-point representative subset models residuals around a fitted linear mean. The original third-pulse holdout remains outside GP fitting. However, the 10% full-test traces now participate in

penalty-weight selection; they are tuning results, not independent validation. Five-percent data are never used to choose the policy in this study.

3 Step 1: the deployable controller

For standardized inverse-model inputs $z_j = (v_j - \bar{v}_j)/s_j$ with $v = [N, \dot{N}, T, P]^\top$, retain the full quadratic

$$p_2(v) = c_0 + \sum_{j=1}^{4} c_j z_j + \sum_{j=1}^{4} c_{jj} z_j^2 + \sum_{i<j} c_{ij} z_i z_j. \tag{1}$$

At each sample, the actual control law is

$$u_{\mathrm{ff},k} = p_2(r_k, 0, T_k, P_k), \qquad e_k = r_k - N_{s,k}, \tag{2}$$

$$u_k^* = u_{\mathrm{ff},k} + K_p e_k + I_k, \qquad u_k = \mathrm{clip}(u_k^*, 0.2, 4), \tag{3}$$

$$I_{k+1} = I_k + hK_i e_k \quad \text{if conditional integration permits it.} \tag{4}$$

The integral updates if ($u^* < 4$ or $e < 0$) and ($u^* > 0.2$ or $e > 0$), allowing correction out of saturation. Before 45 s, the old command drives the plant and the new integral tracks $I = u_{\mathrm{old}} - u_{\mathrm{ff}} - K_p e$ for bumpless transfer. Gains have units (lbm/s)/rpm and (lbm/s)/(rpm s).

Only K_p and K_i are learned. The polynomial's fifteen coefficients are fixed. At this single setpoint and ambient condition, feedforward is constant and an integral offset can compensate its value. The experiment therefore assesses PI tuning, not the benefit of the quadratic surface across speeds. The setpoint remains slightly below the old polynomial training-speed minimum of 9264.23 rpm.

4 Step 2: distinguish magnitude, slew, total variation, and ringing

Fuel magnitude describes how much fuel is commanded. Slew is the local rate $(u_k - u_{k-1})/h$. Total variation counts accumulated command motion, while excess variation removes the unavoidable net motion between endpoints:

$$\mathrm{TV}_{a:b}(u) = \sum_{k=a+1}^{b} |u_k - u_{k-1}|, \tag{5}$$

$$R_{a:b}(u) = \mathrm{TV}_{a:b}(u) - |u_b - u_a| \geq 0. \tag{6}$$

The inequality follows from the triangle inequality. For a monotone trajectory $[0, 0.5, 1]$, TV and net change both equal 1, so ringing is zero. For $[0, 1, 0, 1]$, TV is 3 and net change is 1, so ringing is 2. A direct excess-TV penalty discourages reversals without penalizing a required monotone fuel transition as strongly as plain TV would.

The metric has units lbm/s because it sums changes in a fuel-flow command. It is neither integrated fuel consumption nor the integral of squared fuel slew. It is sensitive to sample rate and noise. This study keeps the sample rate fixed for every comparison. Penalizing ringing does not automatically bound peak slew: the report separately measures that quantity.

5 Step 3: augment the state to preserve temporal correlation

Use normalized physical and controller states

$$x = \begin{bmatrix} (N - 9250)/20 \\ (N - N_s)/5 \\ (I - I_0)/0.3 \end{bmatrix}, \quad I_0 = u_0 - u_{\mathrm{ff}}, \quad \tilde{u} = \frac{u - u_0}{0.3} = b^\top x, \tag{7}$$

where $u_0 = 1.39529998$ lbm/s and $b^\top = [-20K_p/0.3,\ 5K_p/0.3,\ 1]$ for the unsaturated fixed-reference policy. To evaluate fuel increments and endpoint changes, introduce

$$\xi = \begin{bmatrix} x \\ p \\ a \end{bmatrix}, \qquad p = \tilde{u}_{k-1}, \qquad a = \tilde{u}_{\text{event start}}, \qquad \bar{b} = \begin{bmatrix} b \\ 0 \\ 0 \end{bmatrix}. \tag{8}$$

The five-dimensional covariance includes the cross-covariance between current state, previous command and event-start anchor. Ignoring these correlations would generally overestimate incremental-command uncertainty. In particular,

$$\mathrm{Var}(\tilde{u}_k - p_k) = \mathrm{Var}(\tilde{u}_k) + \mathrm{Var}(p_k) - 2\,\mathrm{Cov}(\tilde{u}_k, p_k). \tag{9}$$

At the beginning of each planning event, set the anchor to the current policy output. This is a deterministic linear transformation $\xi \leftarrow T\xi$, where T is identity except its fifth row is $\bar{b}^\top$. Thus both moments transform: $m \leftarrow Tm$ and $S \leftarrow TST^\top$. Resetting only the anchor mean would be incorrect.

6 Step 4: propagate the unchanged GP and augmented moments

The normalized GP input is

$$z = [x_1, x_2, \tilde{u}, d]^\top = B\xi + c(d), \qquad d = P_{\text{load}}/1537.6791946. \tag{10}$$

With $\xi \sim \mathcal{N}(m, S)$, its input moments and state/input cross-covariance are $m_z = Bm + c$, $S_z = BSB^\top$, and $C_{\xi z} = SB^\top$. The GP returns two increments $y = [\Delta x_1, \Delta x_2]^\top$. Integration and command memory are known transitions:

$$\xi^+ = D\xi + Ey, \qquad D = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ -20hK_i/0.3 & 5hK_i/0.3 & 1 & 0 & 0 \\ -20K_p/0.3 & 5K_p/0.3 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}, \qquad E = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}. \tag{11}$$

The fourth state becomes the current command, and the fifth retains the event anchor. The GP residual uses a shared squared-exponential kernel,

$$k(z, z') = \sigma_f^2 \exp[-\tfrac{1}{2}(z - z')^\top \Lambda^{-1}(z - z')], \qquad K_n = K + \sigma_n^2 \mathcal{I} + 10^{-9}\mathcal{I}. \tag{12}$$

For Gaussian input, let $q_i = \mathbb{E}[k(z, z_i)]$, $Q_{ij} = \mathbb{E}[k(z, z_i)k(z, z_j)]$ and $\beta = K_n^{-1}R$, where R contains standardized training residuals. Then

$$q_i = \sigma_f^2 \frac{|\Lambda|^{1/2}}{|S_z + \Lambda|^{1/2}} \exp[-\tfrac{1}{2}(z_i - m_z)^\top (S_z + \Lambda)^{-1}(z_i - m_z)], \tag{13}$$

$$\mu_g = \beta^\top q, \tag{14}$$

$$V_g = \beta^\top Q\beta - \mu_g\mu_g^\top + \mathcal{I}[\sigma_f^2 - \mathrm{tr}(K_n^{-1}Q) + \sigma_n^2]. \tag{15}$$

The two-kernel integral Q uses the midpoint $(z_i + z_j)/2$ and covariance $\Lambda/2$; the full expression and previous validation are available in the preceding derivation paper. Restoring residual output scales and the fitted linear mean yields $\mu_y, V_y, C_{\xi y}$. The recursion is

$$m^+ = Dm + E\mu_y, \tag{16}$$

$$S^+ = DSD^\top + EV_yE^\top + DC_{\xi y}E^\top + EC_{\xi y}^\top D^\top. \tag{17}$$

Each propagated state is approximated as Gaussian. GP noise is treated as independent at successive transitions. This is uncertainty propagation under a model approximation, not a probabilistic safety guarantee or a temporally coherent GP function draw.

7 Step 5: derive expected absolute command motion

For a scalar Gaussian $Z \sim \mathcal{N}(\mu, v)$, split the integral at zero:

$$\mathbb{E}|Z| = \int_{-\infty}^{0} (-z)\mathcal{N}(z \mid \mu, v)\, dz + \int_{0}^{\infty} z\mathcal{N}(z \mid \mu, v)\, dz \tag{18}$$

$$= \sqrt{\frac{2v}{\pi}} \exp\left(-\frac{\mu^2}{2v}\right) + \mu \operatorname{erf}\left(\frac{\mu}{\sqrt{2v}}\right) \equiv F(\mu, v). \tag{19}$$

For zero variance, take the continuous limit $F(\mu, 0) = |\mu|$. For zero mean and positive variance, $F(0, v) = \sqrt{2v/\pi}$, which is nonzero. This exposes why penalizing only $|\mathbb{E}[\Delta u]|$ would miss uncertain command motion.

Define $w_\Delta = \bar{b} - e_4$ and $w_A = \bar{b} - e_5$. After a transition,

$$\mathbb{E}|\tilde{u}_k - \tilde{u}_{k-1}| = F(w_\Delta^\top m_k, w_\Delta^\top S_k w_\Delta), \tag{20}$$

$$\mathbb{E}|\tilde{u}_b - \tilde{u}_a| = F(w_A^\top m_b, w_A^\top S_b w_A). \tag{21}$$

Linearity of expectation permits summing absolute-increment expectations even when increments are correlated. For each event window,

$$\mathcal{R}_{a:b} = \sum_{k=a+1}^{b} F(w_\Delta^\top m_k, w_\Delta^\top S_k w_\Delta) - F(w_A^\top m_b, w_A^\top S_b w_A). \tag{22}$$

The corresponding physical expected ringing is $0.3\mathcal{R}$ lbm/s. No extra factor of h belongs in this TV sum: $h\sum|\Delta u/h| = \sum|\Delta u|$. An independent Monte Carlo test checks the folded-normal formula, and a particle rollout cross-checks the complete augmented calculation on the same GP.

8 Step 6: form the objective and search penalty weights

The former tracking and magnitude costs are retained:

$$J_{\text{track}} = h\sum_k (m_{1,k}^2 + S_{11,k}), \tag{23}$$

$$J_{\text{fuel}} = 0.01h\sum_k [(\bar{b}^\top m_k)^2 + \bar{b}^\top S_k \bar{b}], \tag{24}$$

$$J_\lambda = J_{\text{track}} + J_{\text{fuel}} + \lambda \sum_{\text{events}} \mathcal{R}_{a:b}. \tag{25}$$

The 5 s planning trajectory applies a unit load for $1 \le t < 2.5$ s. On the 15 ms grid, transitions begin at 1.005 and 2.505 s. The load-on event closes immediately before unloading, and the unloading event closes at the end of the horizon. Its event windows differ from the fixed two-second windows used in the plant metric. This difference is declared; the plant metric is never replaced by a model prediction.

The coarse weights are $\lambda \in \{0, 0.001, 0.003, 0.01, 0.03, 0.1, 0.3\}$. Candidate zero retains the previous tracking-only gains. Positive weights use bounded L-BFGS-B on $[\log K_p, \log K_i]$, warm-started from the preceding weight. After inspecting the coarse 10% tradeoff, one refinement adds weights 0.15, 0.20 and 0.25 in the 0.1–0.3 bracket, warm-started from the nearest available weight. This refinement uses tuning data only and is recorded before the added simulations; it does not use 5% validation. Bounds remain $0.005 \le K_p \le 0.07$ and $0.01 \le K_i \le 0.25$. Policy gradients are finite differences of the full probabilistic rollout. This finite set and local optimization do not establish a global optimum.

For each weight, the resulting gains are evaluated in the complete nonlinear 10% test. The selection score, declared before these evaluations, is

$$S_j = \max\left(\frac{\text{RMSE}_j}{2.4209039065}, \frac{R_j}{2.0702280417}\right). \tag{26}$$

The denominators are the historical base PI metrics. The smallest score among numerically accepted candidates wins; the score gives equal priority to the worse of these two relative metrics. It does not minimize settling time, peak error, surge margin, or fuel slew. Gains are frozen before 5% validation.

λ	K_p	K_i	RMSE	Ringing	Score
0	0.06461	0.17103	1.1973	14.8252	7.161
0.001	0.06206	0.16260	1.2054	13.0728	6.315
0.003	0.05751	0.15035	1.2343	10.6050	5.123
0.01	0.04640	0.13319	1.3609	6.8917	3.329
0.03	0.03820	0.10339	1.5716	4.6845	2.263
0.1	0.02949	0.07057	1.9952	2.8858	1.394
0.15	0.02643	0.06097	2.2146	2.3771	1.148
0.2⋆	0.02480	0.05386	2.3877	2.0838	1.007
0.25	0.02355	0.04890	2.5381	1.8762	1.048
0.3	0.02216	0.04512	2.6968	1.6841	1.114

Table 1: All tested penalty weights. RMSE is in rpm; ringing is in lbm/s. The star marks the selected policy.

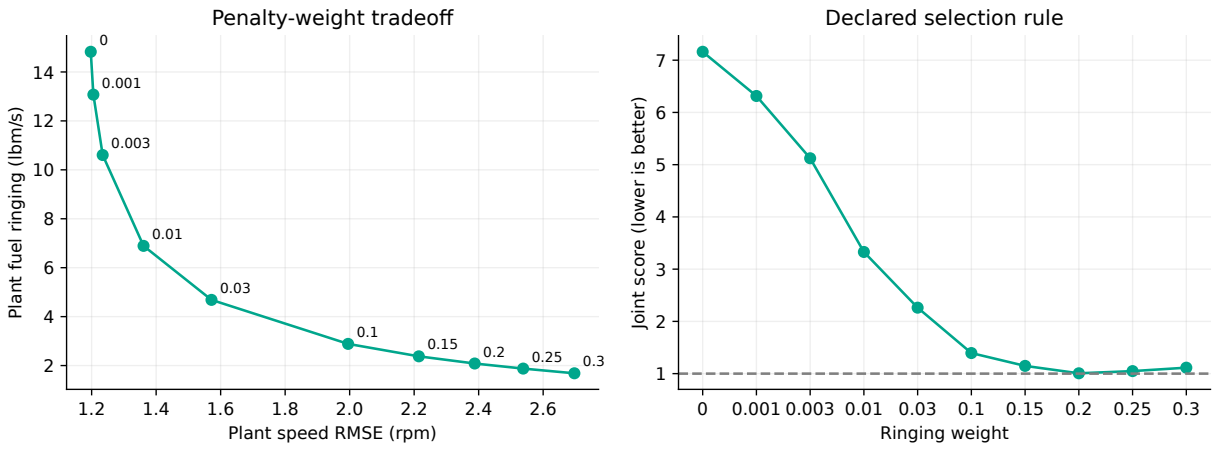


Figure 1: Measured 10% plant tradeoff and the declared selection score. These are candidate evaluations, not independent validation.

9 Step 7: nonlinear results and interpretation

The deployed gains are frozen at $K_p = 0.02479762$, $K_i = 0.05385920$ and $K_d = 0$.

Controller	RMSE (rpm)	Peak (rpm)	Ringing	Settling (s)
Base PI	2.4209	21.406	2.0702	2.475
Original Poly2	1.6658	14.113	1.6968	1.365
Retuned Poly2	1.5461	15.825	1.2990	0.930
Previous PILCO	1.1973	14.389	14.8252	1.920
Smooth PILCO	2.3877	21.593	2.0838	2.220

Table 2: 10% load test. Ringing units are lbm/s; settling uses the persistent 0.1 rpm band.

Controller	RMSE (rpm)	Peak (rpm)	Ringing	Settling (s)
Base PI	1.1692	10.392	1.0583	2.115
Original Poly2	0.8018	6.969	0.9005	1.260
Retuned Poly2	0.7439	7.806	0.6544	0.870
Previous PILCO	0.5930	7.008	7.9471	1.695
Smooth PILCO	1.1533	10.486	1.0615	1.905

Table 3: 5% load test. Ringing units are lbm/s; settling uses the persistent 0.1 rpm band.

Relative to Base PI, smooth PILCO changes RMSE by -1.4%, ringing by +0.7%, and fine settling from 2.475 to 2.220 s.

Relative to Retuned Poly2, smooth PILCO changes RMSE by +54.4%, ringing by +60.4%, and fine settling from 0.930 to 2.220 s.

Relative to Previous PILCO, smooth PILCO changes RMSE by +99.4%, ringing by -85.9%, and fine settling from 1.920 to 2.220 s.

Its sampled peak fuel slew is 4.763 lbm/s^2, versus 10.514 for previous PILCO. Minimum surge margin changes from 14.800% to 18.201%; the selected 10% run has 0.000% fuel saturation. These diagnostics complement the objective rather than entering its selection score.

With the gains frozen, the 5% test gives RMSE 1.1533 rpm and ringing 1.0615 lbm/s, a 86.6% ringing reduction from previous PILCO. This supports amplitude transfer of the smoothing effect at this operating point.

What the video compares. Four synchronized windows show base PI, retuned Poly2 + PID, previous PILCO, and smooth PILCO. Original Poly2 remains in the report tables but is omitted from the four-window video to make room for the new method. Each window has its own speed-error and fuel plots, matching axes, current speed, current fuel, a shaft-load indicator and full-test metrics. Orange shading marks load-on intervals. The video includes the full 10% test, slowed load application/removal, frozen-gain 5% validation and a final scorecard. All plotted values are recorded plant signals without visual smoothing.

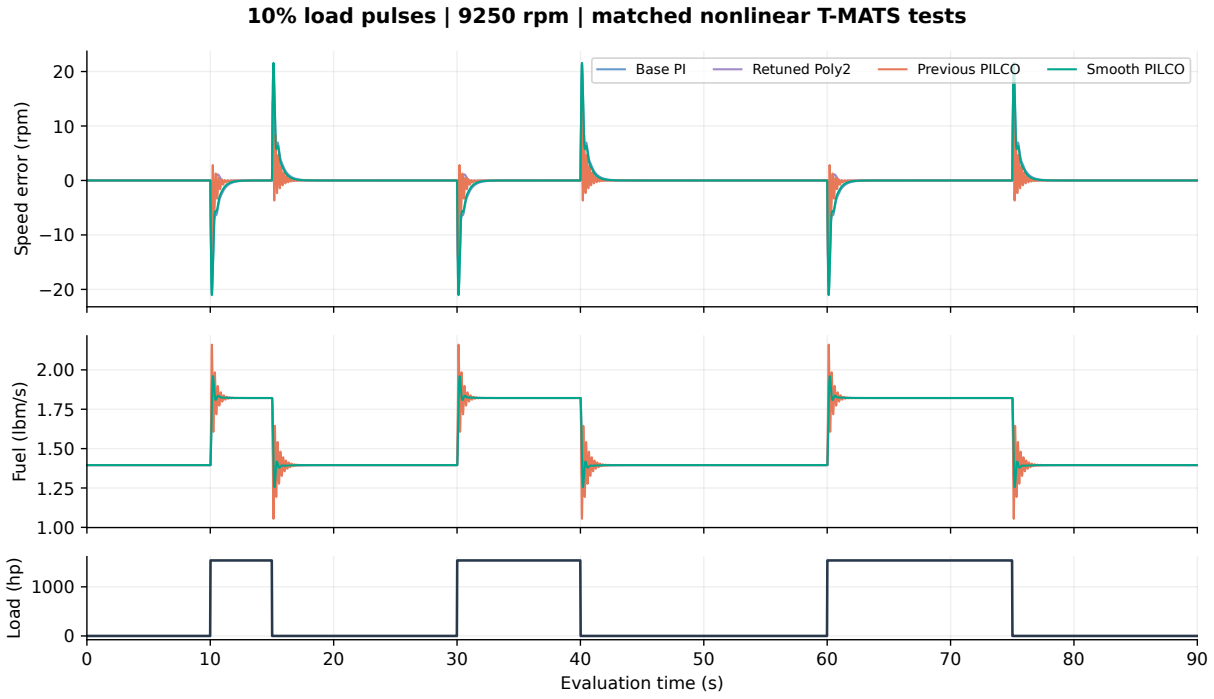


Figure 2: 10% full test; only the controller gains differ between the two PILCO policies.

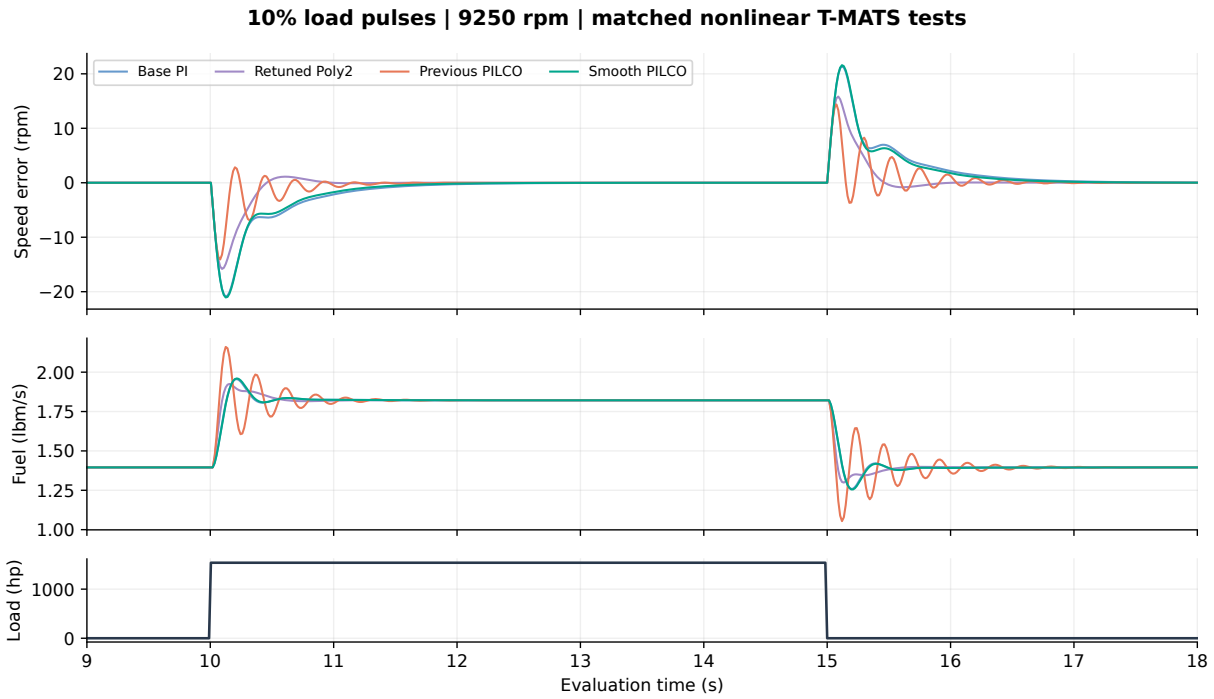


Figure 3: First load pulse, showing the effect of the ringing penalty on physical speed and fuel command.

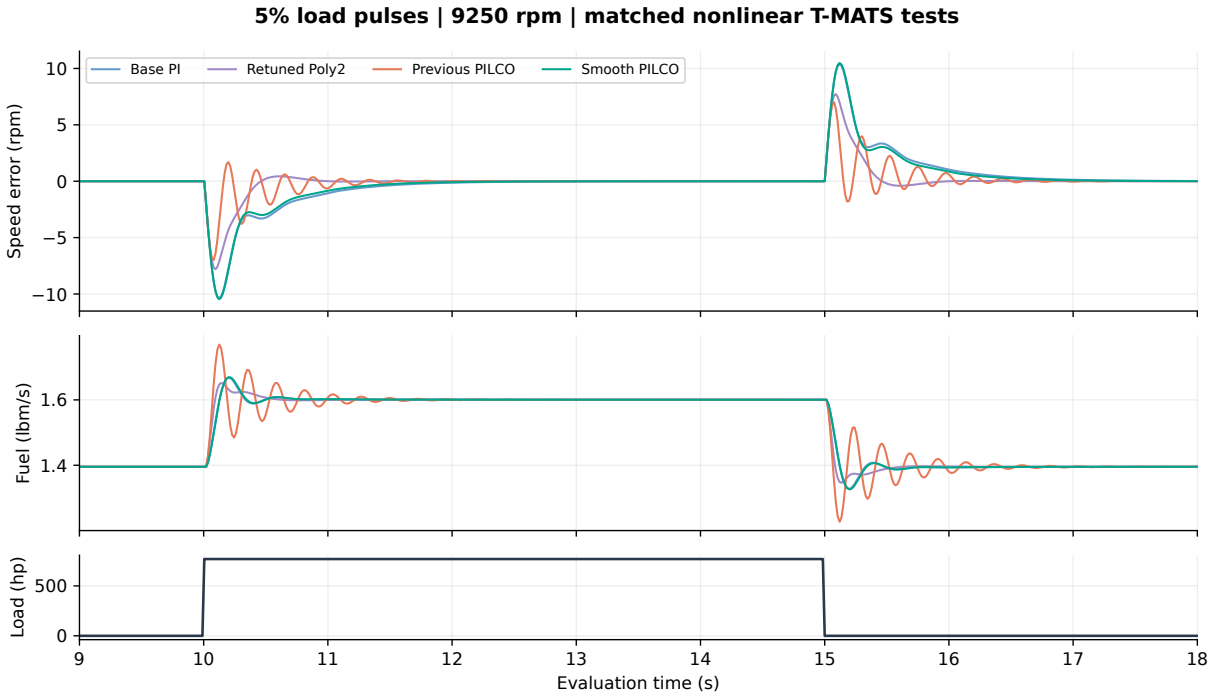


Figure 4: Frozen-policy 5% amplitude validation with the same timing and plant.

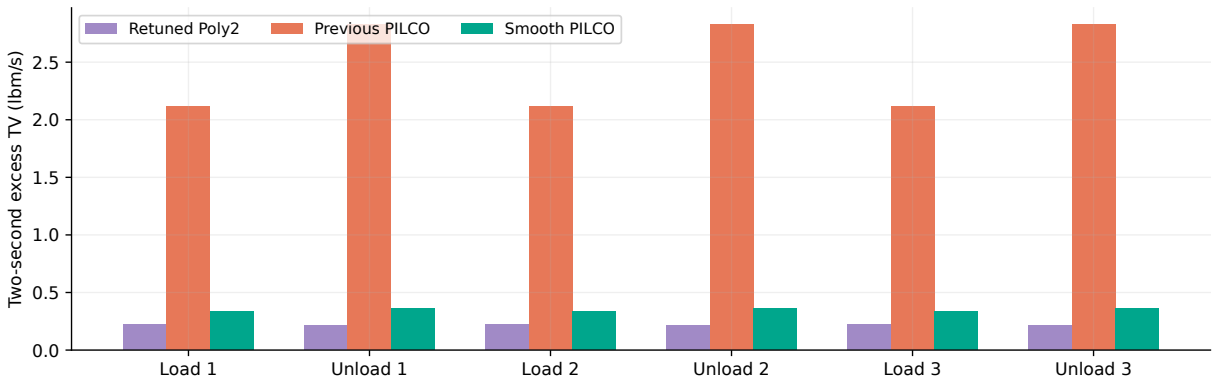


Figure 5: Ringing decomposed into six individual 10% load edges. Each bar is total fuel movement minus net movement in the same two-second window.

10 Step 8: numerical checks and metric definitions

All selected traces pass the shared plant acceptance checks. Maximum independent command replay discrepancy is 1.46e-13 lbm/s; maximum integral update discrepancy is 5.38e-15 lbm/s. The folded-normal formula passes 250,000-sample scalar Monte Carlo checks, including zero-variance and nonzero-variance cases. For the selected policy, a 5000-particle GP rollout predicts normalized excess TV 2.317963, compared with 2.317930 from Gaussian moment closure; their relative difference is 0.0014%. The particle standard error is 0.000037. This checks prediction consistency on the same surrogate; it does not independently validate the physical GP uncertainty.

The plant acceptance test requires finite signals, positive fuel, positive surge margin, flow residual at most 10^{-9}, fewer than 200 solver iterations, full 150 s duration and preparation speed within 0.1 rpm. Speed RMSE and peak error use all evaluation samples. IAE uses the previous study's rectangular sum. Ringing sums TV minus endpoint motion in six two-second load-edge windows. Fine settling requires staying inside ± 0.1 rpm until the next event; it is not a broad percentage-of-speed settling criterion. Reported peak slew is the maximum sampled fuel change divided by 0.015 s in those windows.

Independent replay reconstructs $u^* = u_{\mathrm{ff}} + K_p(r - N_s) + I$ and every eligible integral update, then checks the saved command. Metrics are recomputed from CSV and compared against MATLAB's shared implementation. Identical time, reference and load vectors are checked across compared runs. The extra previous-command and anchor states exist only in offline prediction: the runtime controller still requires just its integral state.

11 Limits and how to use the result

The result is smoother than the earlier tracking-only PILCO, but the complete comparison must also account for tracking, settling and surge margin. It does not dominate the retuned Poly2 controller on both speed RMSE and ringing. That comparator has derivative/acceleration feedback, whereas the new policy is restricted to PI. The chosen weight is the best under the declared score among the tested candidates, not a universal tuning value.

This is an offline PILCO-style method, not a reproduction of the original full collect–refit–improve loop. The original PILCO work motivates GP dynamics and uncertainty-aware policy evaluation [1]. This implementation uses a fitted linear mean, a subset GP with shared covariance hyperparameters, a restricted PI policy, numerical gradients, and a custom expected excess-TV cost. It assumes an unsaturated affine policy during planning; plant runs apply the actual saturation and anti-windup. Five-percent testing checks amplitude transfer at one operating point and is not proof of robustness to noise, speed changes, ambient changes or hardware dynamics.

To prioritize still less actuator activity, select a larger penalty only after examining its tracking cost. To require a hard slew bound, add and validate an explicit actuator-rate constraint or a different policy architecture; an excess-TV penalty alone does not provide such a bound. To claim advantages for the quadratic feedforward map itself, perform separate multi-speed and ambient validation.

12 Reproduction

Run `pilco_smooth.py` to load the frozen GP and optimize the coarse weights. Run `run_tmats_pilco_smooth('tune')` in MATLAB. Run `refine_pilco_smooth.py`, then repeat the MATLAB tuning call, which resumes saved candidates and evaluates the additions. Run `select_pilco_smooth.py` to write `selected_policy.json`, then `run_tmats_pilco_smooth('validate')` only after selection. The ready-to-open model is `GasTurbine_PILCO_Smooth_PI.mdl`; its preload callback loads the frozen selected policy. Run `check_pilco_smooth_mc.py` for the particle cross-check and the report/video builders for final artifacts. Outputs are in `results/pilco_smooth_20261010`. This is an overlay for the existing T-MATS workspace and adjacent library; earlier studies remain unchanged.

References

[1] M. P. Deisenroth and C. E. Rasmussen. PILCO: A Model-Based and Data-Efficient Approach to Policy Search. ICML, 465–472, 2011. `https://icml.cc/2011/papers/323_icmlpaper.pdf`.

[2] Supplied workspace. Previous PILCO-style quadratic feedforward + PI study and fixed-speed Poly2 study, `results/pilco_quadratic_pi_20261010` and `results/poly2_tune9250_20260915`. Prior derivation: `output/pdf/PILCO_Quadratic_FF_PI_Comparison.pdf`.